Synopsis

END TO END PRE-LOAN DEFAULT ASSESSMENT SYSTEM

Problem Id: 2

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Problem Description

The financial sector faces significant risks from "credit risk," where borrowers fail to repay loans, leading to substantial economic losses. This project aims to develop a predictive model that identifies potential "loan defaulters" based on historical customer banking data. By leveraging classification algorithms, banks can automate the risk assessment process, ensuring more objective and data-driven lending decisions. In real life, this minimizes non-performing assets (NPAs) and helps maintain the stability of the banking ecosystem.

Steps of Implementation

1. Exploratory Data Analysis (EDA): Analyzing data distributions and identifying patterns or correlations using heatmaps and scatter plots.

2. Data Preprocessing: Cleaning missing values, normalizing numerical features, and performing categorical encoding.

3. Dimensionality Reduction (PCA): Applying Principal Component Analysis to reduce the feature space while retaining maximum variance, optimizing model efficiency.

4. Model Selection & Training: Implementing Support Vector Machines (SVM) with various kernels (Linear, Poly, RBF) and K-Nearest Neighbors (K-NN).

5. Evaluation & Validation: Comparing models using performance metrics to select the most robust classifier for final deployment.

STEPS TO IMPLEMENT ML SYSTEM FOR PREDICTING LOAN DEFAULT

Objective: Predict whether a customer will **Default** on a loan

EXPLORATORY DATA ANALYSIS (EDA)

Analyze distribution, correlations, and patterns in banking data

- Visualize distributions
- Identify correlations
- Spot anomalies

DATA PREPROCESSING

Clean and preprocess data for modeling

- Handle missing values
- Normalize and transform data
- Reduce dimensionality with PCA

TRAIN-TEST SPLIT

70% 30%

Divide data into training and testing sets

- Typically 70% training / 30% testing

MODEL SELECTION

Implement and compare classification models.

- SVM (Linear, Polynomial, RBF Kernel)

K-NN

MODEL TRAINING

Train the selected models on the training data using optimal hyperparameters

MODEL EVALUATION

Evaluate and compare the models using metrics.

Precision | Recall | F1-Score | ROC-AUC

EXPECTED OUTPUT

1 Default

0 No Default

Expected Modules / Libraries to be used

• Pandas: Used for data manipulation, cleaning, and managing structured data in DataFrames.

• NumPy: Provides support for high-performance mathematical operations on multi-dimensional arrays.

• Scikit-Learn: The core library for implementing PCA, SVM, K-NN, and various evaluation metrics.

• Matplotlib / Seaborn: Essential for generating visual plots like ROC curves and correlation matrices. Used for creating static, animated, and interactive visualizations to understand data distribution and model results.

Expected Evaluation Metrics and Plots to be used

• Precision & Recall: To measure the accuracy of positive predictions and the model's ability to find all actual defaulters.

• F1-Score: Provides a balanced harmonic mean between precision and recall, crucial for imbalanced banking data.

• ROC–AUC Curve: To visualize the trade-off between the true positive rate and false positive rate across different thresholds.

• Confusion Matrix: A plot used to visualize the performance of the classification algorithm and identify misclassifications.

Required Topics from the Subject

1. Applied Statistics & EDA

Central Tendency: Mean, Median, and Mode for filling missing data.
Outlier Detection: Using the Interquartile Range (IQR) to identify and remove extreme data points.
Visualization: Understanding Correlation Heatmaps to see how features (like Income) relate to the target (Loan Default).

2. Data Preprocessing

Imputation: Techniques for handling missing values (NaNs).
Encoding: Converting categorical text into numbers (One-Hot and Label Encoding).
Feature Scaling: Standardizing data (Standard Scaler) so that features with large ranges don't dominate the model.

3. Dimensionality Reduction (PCA)

Feature Space: Understanding why too many variables can lead to "Overfitting."
Principal Component Analysis: The math of compressing many features into a few "Principal Components" while keeping the most important information.

4. Classification Algorithms

K-Nearest Neighbors (KNN): Predicting based on the similarity of "neighboring" data points.
Support Vector Machines (SVM): Using the "Kernel Trick" (Linear, Poly, RBF) to find a boundary that separates classes.

5. Evaluation Metrics

Confusion Matrix: A table showing True Positives vs. False Negatives.
Accuracy vs. Recall: Understanding why "Recall" is critical in banking (catching defaulters) even if overall accuracy is high.
F1-Score: The harmonic mean that balances Precision and Recall for a more honest performance score.

Platform Required:

- Platform: Jupyter Notebook / Python IDLE.
- Language: Python 3.x

Books and Link Sources

1. Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow by Aurélien Géron.

2. Pattern Recognition and Machine Learning by Christopher Bishop.

3. Scikit-Learn Official Documentation (scikit-learn.org).

4. Kaggle Datasets and Tutorials for Credit Risk Modeling.